

Switch model

Thursday, August 7, 2025

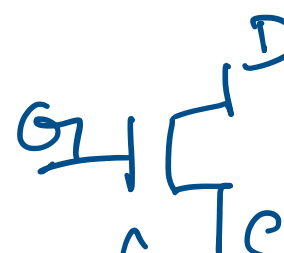
4:06 PM

Switch model of logic gates



Pull up network: pulls the output "up" to logic 1

Pull down network: pulls the output "down" to logic 0

Switch $\equiv$ nMOS 

Comp. switch $\equiv$ pMOS 

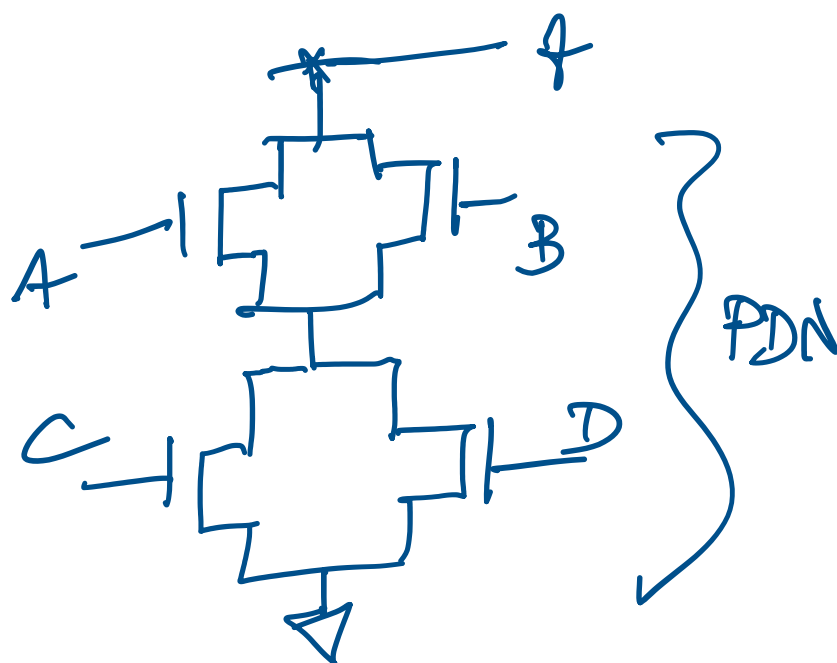
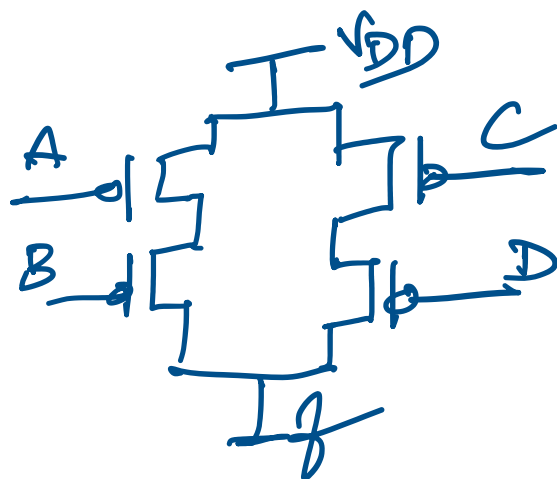
PUN and PDN are duals of each other.

PDN - nMOS network

PUN - pMOS network

e.g. $f = (A+B) \cdot (C+D)$

PUN



HW

Try $f_1 = (A+B) \cdot C$

$f_2 = AB + C + D$